Introduction

Whether we are using email, social media, banking services or enterprise networks, passwords are still the most prevalent way to verify users' identities in digital systems. Even with the advent of biometrics, hardware tokens and multi-factor authentication, password-based authentication remains the most common type of authentication because it is both low-cost and easy to implement and use. The strength of a password is the first line of defense against a user's data being accessed by a potential attacker and they will be more resistant to guess, brute force, and dictionary attacks the stronger that password is.

A "strong" password is a password that an attacker cannot easily guess or that cannot be practically broken by computer in a reasonable amount of time; this is usually accomplished by being long, unpredictable, and random. A weak password, on the other hand, is a short, easy-to-remember password or something that's commonly used, like a dictionary word, a keyboard combination (e.g., "asdfg123") or a previously leaked password. Using weak passwords may lead to a breach of the protected information within seconds to minutes with the help of automated tools, giving other parties access to sensitive information.

The traditional password strength checkers, which are used by most websites and applications today, use static, rule-based heuristics. These are usually a minimum of a certain number of characters and a combination of characters, numbers, and symbols (for example, both lowercase letters and digits and special symbols). It's easy to do but is essentially flawed in that it checks on the password's composition, not its actual unpredictability. However, a password like " Password1!" meets all the rules-based criteria and could be considered “strong”, even though it is one of the most frequently used and guessed passwords in leaked database of credentials. This leads to a potentially hazardous difference between a user's perception of security and the real security, and thus endows users with false sense of security.

An alternative to that problem, and an approach that is data-driven is machine learning (ML). Instead of static rules, ML models are trained on large collections of real or leaked passwords to learn patterns that can be used to distinguish between truly strong passwords and weak ones, which can include things such as substituting characters in a password ("P@ssw0rd") or adding predictable suffixes ("123", "!") or closeness to common or previously-leaked passwords. Machines can learn from data instead of using hard-coded rules, potentially improving over the rule-based systems widely used today by providing a more realistic estimate of the crackability of the password in the real world.

The purpose of this research is to create machine learning method for precise detection and classification of the strength of the password.

Objectives

- Examine past research and review existing rule-based password meters, password strength estimation and classification techniques.
- Create a Machine Learning-based password strength classification model with appropriate extracted characteristics (character composition, pattern similarity, dictionary similarity, length and entropy).
- Implement the design outlined in objective (2) above.
- Test the proposed system with common metrics: accuracy, precision, recall and F1-score and compare the results with other rule-based password strength meters.

Tools and Technologies

Category	**Tool**	**Purpose**
Programming Language	Python	Core implementation language
ML Library	scikit-learn	Random Forest / SVM classifier training and evaluation
Data Handling	pandas, numpy	Dataset loading, cleaning, feature engineering
Visualization	matplotlib	Result charts (accuracy, confusion matrix, comparisons)
Environment	Jupyter Notebook	Development and experimentation
Dataset Source	Public leaked-password corpora (e.g., RockYou) or synthetic dataset	Training and testing data
Version Control	GitHub	Code tracking